

Study Questions: Iterative Refinement

Analytical Questions

1. Describe the action-perception loop in prototyping. How does each turn of the loop reduce free energy?
2. Why does Active Inference predict that many small, fast iterations are generally more valuable than a few large, slow ones? What is the relationship between loop frequency and model convergence?
3. Define the minimum viable prototype in Active Inference terms. What is being maximized and what is being minimized?
4. Explain how A/B testing functions as controlled action selection. What is the Bayesian rationale for testing both options rather than choosing one based on intuition?
5. Distinguish between "action within a model" (persevering) and "model switching" (pivoting). What signals indicate when each is appropriate?
6. How does the expected free energy framework guide the choice of which design change to make next? What are the epistemic and pragmatic components?
7. What is an iteration roadmap, and why does ad hoc iteration tend to become circular rather than convergent without one?
8. Explain how the Toyota Production System's kaizen philosophy implements the Active Inference action-perception loop at an organizational scale.
9. What is the relationship between prototype fidelity and iteration speed? Why might deliberately reducing fidelity increase the overall value of the iteration process?
10. How does the Stage-Gate product development process enforce disciplined action selection? What is the function of "gates" in the iteration process?

Applied Questions

1. You are developing a new type of reusable water bottle. You have three unresolved design questions: (a) optimal capacity, (b) lid mechanism, (c) insulation material. In what order should you test these, and why? Design the first three iterations of your roadmap.
2. A startup has been iterating on their food delivery app for six months. Monthly active users have grown from 100 to 500 but have plateaued. Apply the pivot-or-persevere framework: what signals should they look for to decide whether to keep iterating or pivot to a different approach?
3. You are choosing between two handle designs for a garden tool. Design an A/B test that would resolve this choice, specifying controlled variables, measurement criteria, sample size, and decision rules.
4. Drew Houston's Dropbox video demonstrated the MVP concept. Identify another invention or product where a non-functional prototype (video, mock-up, or simulation) could test the most critical assumption more efficiently than a functional prototype.
5. SpaceX's approach to Starship development involves intentionally destroying expensive hardware to gather test data. Under what conditions does this strategy minimize expected free energy more effectively than simulation-based development? When would it not?
6. You have completed 8 iterations of your prototype and each iteration has improved performance by approximately 2%. Is this a persevere or pivot signal? What additional information would you need to decide?
7. An inventor is torn between building one high-fidelity prototype or five low-fidelity prototypes with the same budget. Analyze this trade-off using the concepts of loop frequency, information gain per iteration, and expected free energy.
8. Your team is debating whether to invest in automated testing equipment that would allow 10x more test cycles per week. The equipment costs 6 months of salary. Use the expected free energy framework to evaluate this investment.

9. Describe a situation where a pivot turned out to be wrong — the original approach was actually correct but was abandoned too early. What reasoning errors might lead to a premature pivot?
10. You are at iteration v7 of your prototype. v1-v5 showed steady improvement, but v6 and v7 showed slight degradation. Diagnose this pattern using Active Inference concepts and recommend a specific next action.